

ROOTMESH · DESIGN DECISIONS REGISTER

RootMesh design decisions register

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DOCUMENT CONTROL

Document ID	RMS-DEC-001	Version	0.3
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/rootmesh		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Follow-ups of 2026-10-02 carried out; cost restated with the status LED (line 17)
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 7 on 2026-10-02 (RMS-DDR-003 accepted, fin printed on a 300 mm class printer, head bonded, R9 cell limit restated on condition, Texas A&M AgriLife Extension on US915 as first candidate, status light, gateway indoors); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note and decision records, the design for construction (RMS-DDR-003), items to confirm when parts are bought, and the value-engineering summary

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note (`docs/REVIEW.md`); this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The moisture probe boards are 23 mm wide and 98 to 100 mm long, with their parts on one face within 16 x 3 mm at the top end	The fin's slots and electronics pockets are sized for them	RMS-DDR-003, P4
2	The temperature probe's sheath is 6 mm across and about 50 mm long, and its inner wires fit a 3.5 x 5 mm channel with the jacket stripped	The fin's bore and channel are sized for it	RMS-DDR-003, P5
3	The 14500 cell with its fuse and leads fits a 15 mm bore, 51 mm deep	The cell holder is sized for it	RMS-DDR-003, P7
4	The PVC pipe is 42 mm outside and 36 mm inside diameter	The head socket (43 mm) and fin spigot (35.4 mm) leave glue gaps of about 0.5 and 0.3 mm	RMS-DDR-003, P3
5	A 64 x 2 mm O-ring, an M8 gland for 2 to 4 mm cable with its locknut, and short M3 heat-set inserts for a 4.6 mm hole are available	The cap groove, gland boss and insert bosses are sized for them	RMS-DDR-003, P1 and P8
6	The solar panel is 70 x 50 mm or smaller, with its lead near the centre of its back	The cap recess is 71 x 51 mm with a lead hole at its centre	RMS-DDR-003, P1
7	The charger's temperature window is 0 to 45 °C	R7 and safety stops S3 to S5 depend on it	RMS-CAL-001, B6
8	The shaft collars fit 20 mm bar	The slot tool handle depends on them	RMS-DDR-003, P11

■ Value engineering

Value-engineering target: USD 300. Estimated cost of the constructable design: USD 281.45 (USD 18.55 under the target), for the pilot set of three stakes, one gateway and one slot tool; USD 59.15 per stake against R12's USD 60. The target is a hypothetical control target, not a limit. Main cost drivers and savings worth trying:

- The largest lines are the gateway (USD 90, about a third of the set), the controller board (USD 15 a stake, USD 45 for three), the antenna and feed (USD 7 a stake) and the seals (USD 5 a stake).

- Making the design constructable added USD 3.50 a stake (the two-part head, the fin halves, and the inserts, screws, epoxy and prototype board of line 16) and USD 4 on the slot tool (shaft collars and thumb screw); the set rose from USD 266.50 to USD 281, and to USD 281.45 on 2026-10-02 with the status LED and resistor of line 17 (USD 0.15 a stake).
- Savings worth trying: one gateway serves a whole farm, so the per-stake cost falls quickly in larger sets; the controller board's price should fall once the carrier board is designed at TRL 4; one slot tool serves every stake on a farm; O-rings, glands and inserts are much cheaper bought by the hundred.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D9: solar with a LiFePO4 cell, antenna at 1 m on the marker rod, The Things Network with an indoor gateway, the budget read as a pilot set, LiFePO4 chemistry, two moisture depths, modified hobby probes, Wio-E5 controller, 20 min interval	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	RMS-DDR-001
2026-09-25	Slot tool for firm soil; deeper cell checked and kept at 62 mm; pilot set includes the slot tool; field calibration note decided but on hold	Amish: "i accept all your recommendations, go with them across all repos."	RMS-DDR-002, A1 to A4
2026-09-30	Build plan format approved for all repos; open decisions go in this register, not in the build plan; the design is made physically buildable as the pictures are drawn	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"; "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	RMS-BLD-001, RMS-DDR-003
2026-10-01	The budget is a value-engineering target, not a limit; budget_usd stays USD 300	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering
2026-10-02	Design for construction accepted, P1 to P11, as recorded (two-part head with O-ring and screws, which also settles the separate solar cap of the 2026-09-26 review; board guides; fin in two glued halves with collar and spigot; flush temperature probe; wedge tip and one-piece electrodes; cell holder; gland and vent moved; antenna clips; slot tool handle and stop fixings)	Amish: "i approve your recommendations for all 555 open decisions."	RMS-DDR-003, P1 to P11
2026-10-02	Fin print size: option (a), each 355 mm fin half is printed on a 300 mm class printer or through a print service	Amish: "i approve your recommendations for all 555 open decisions."	RMS-DDR-003, A1
2026-10-02	Head joint: option (a) for the prototype, the head is bonded to the tube; revisited after the TRL 4 soak test	Amish: "i approve your recommendations for all 555 open decisions."	RMS-DDR-003, A2
2026-10-02	Cell temperature in bare, hot soil: option (B), R9's cell limit restated to the chosen cell's rated discharge range, with charging still blocked above 45 °C, provided its datasheet rates discharge to 55 °C or more; if it does not, the printed shade skirt round the head (option A) is added	Amish: "i approve your recommendations for all 555 open decisions."	RMS-DDR-002, P1
2026-10-02	First users and region: a US university extension program with a research farm, on US915; first	Amish: "i approve your recommendations for all 555 open decisions."	RMS-DDR-001, O1

Date	Decision	Decided by	Record
	candidate to approach Texas A&M AgriLife Extension		
2026-10-02	Status light: keep a green LED on the head, blinking briefly after each uplink and on a magnet or button wake only, once its charge is added to the energy budget (RMS-CAL-001)	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW.md, 2026-09-26, item 2
2026-10-02	Gateway in the hero render: the pilot gateway stays indoors as decided; the post in the hero render is a layout only, and its caption says so	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW.md, 2026-09-26, item 3